

OBJECT ORIENTED PROGRAM

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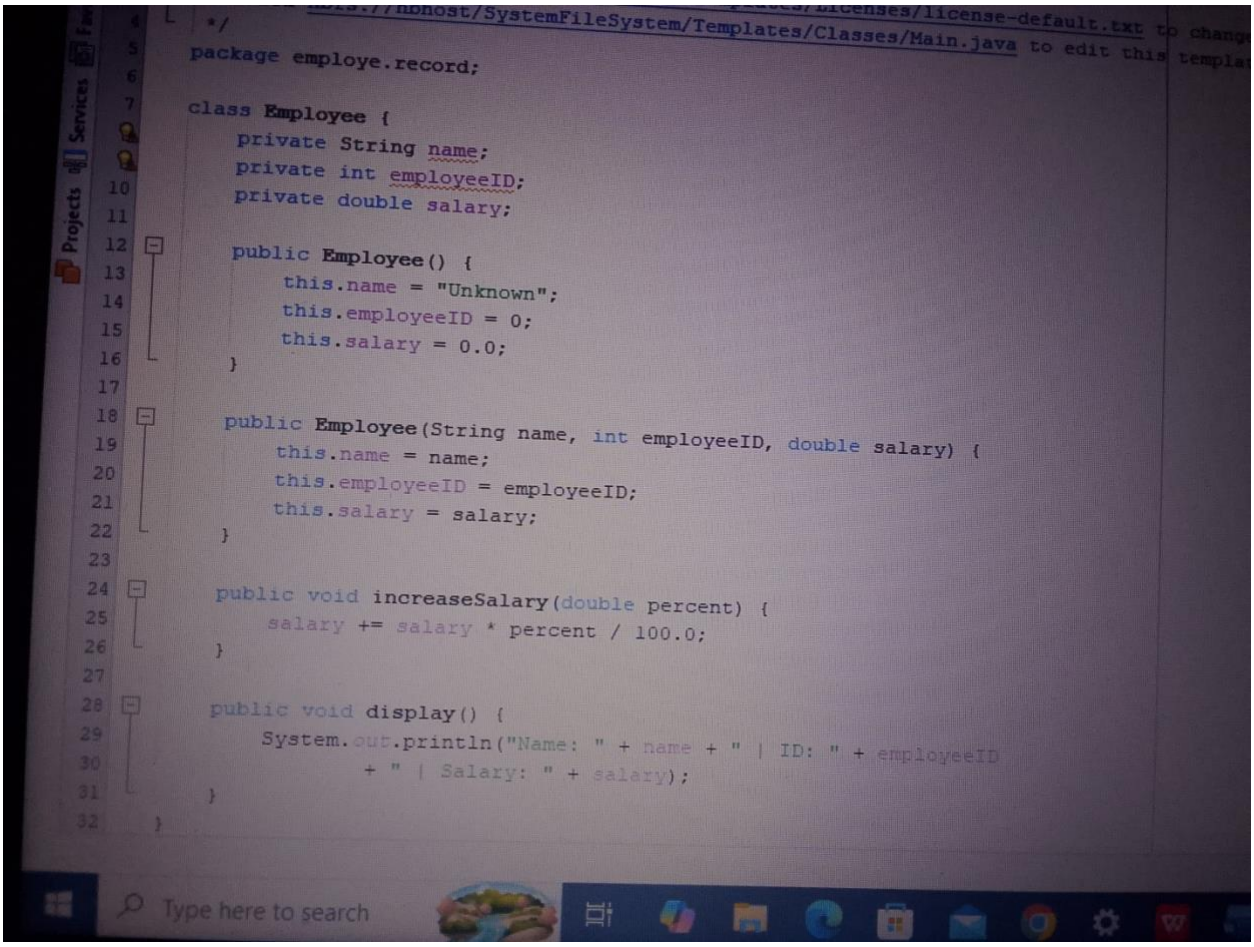
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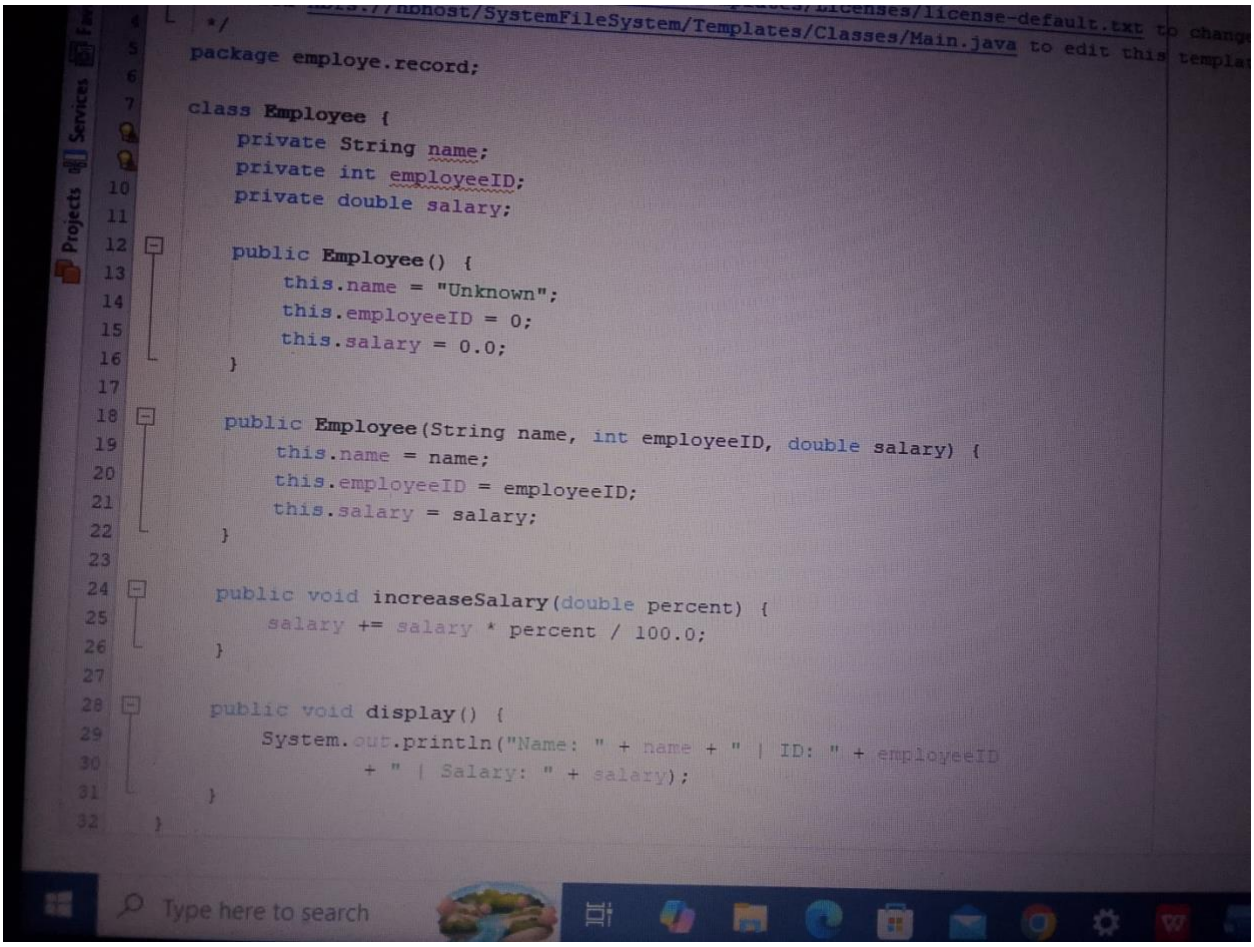
Submitted to: Mam Nayab Khalid

Task 1 — Employee Records

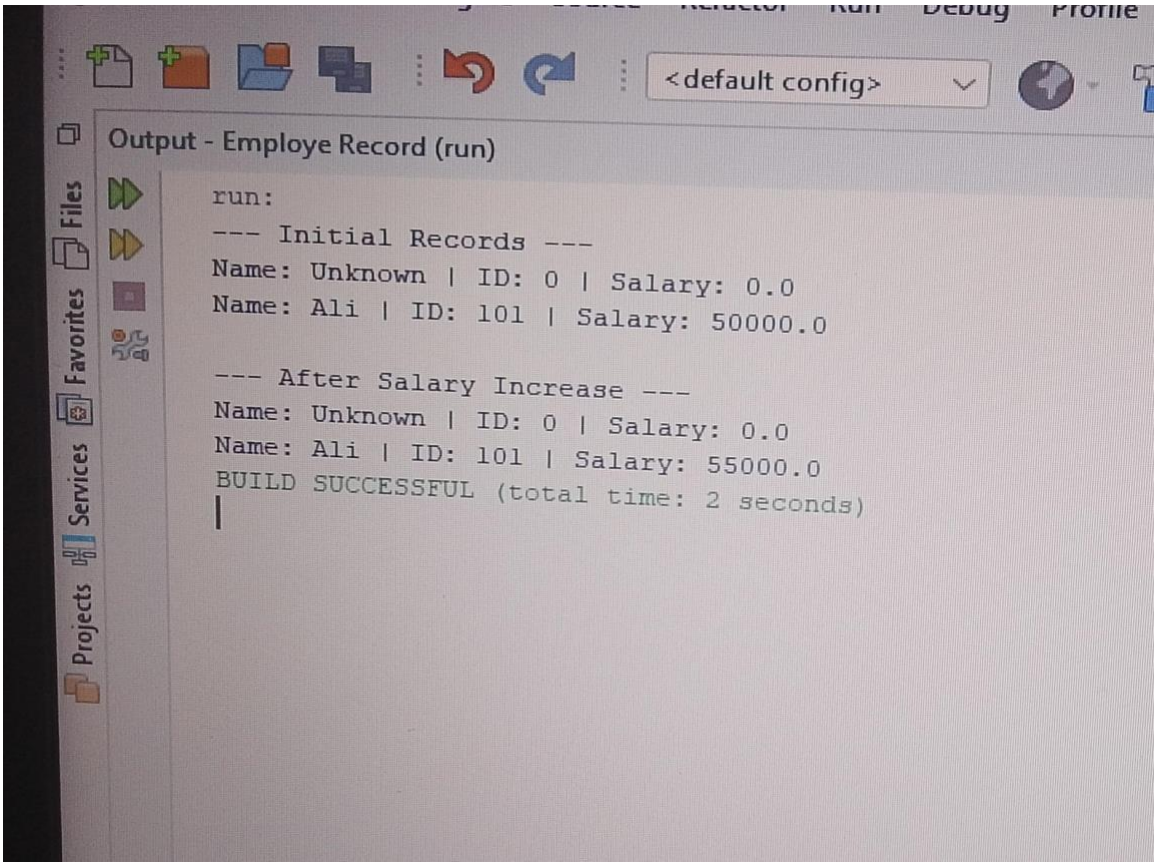
Create an Employee class with private members name, employee ID and salary. Implement both a default constructor and a parameterized constructor. Create two objects and display their information.

Extension: Add a function that increases salary by a given percentage.




*/
package employe.record;
class Employee {
private String name;
private int employeeID;
private double salary;
public Employee() {
this.name = "Unknown";
this.employeeID = 0;
this.salary = 0.0;
}
public Employee(String name, int employeeID, double salary) {
this.name = name;
this.employeeID = employeeID;
this.salary = salary;
}
public void increaseSalary(double percent) {
salary += salary * percent / 100.0;
}
public void display() {
System.out.println("Name: " + name + " | ID: " + employeeID
+ " | Salary: " + salary);
}
}
Type here to search

Output:


Debug
Profile
<default config>
Output - Employe Record (run)
Files
Favorites
Services
Projects
run:
--- Initial Records ---
Name: Unknown | ID: 0 | Salary: 0.0
Name: Ali | ID: 101 | Salary: 50000.0
--- After Salary Increase ---
Name: Unknown | ID: 0 | Salary: 0.0
Name: Ali | ID: 101 | Salary: 55000.0
BUILD SUCCESSFUL (total time: 2 seconds)

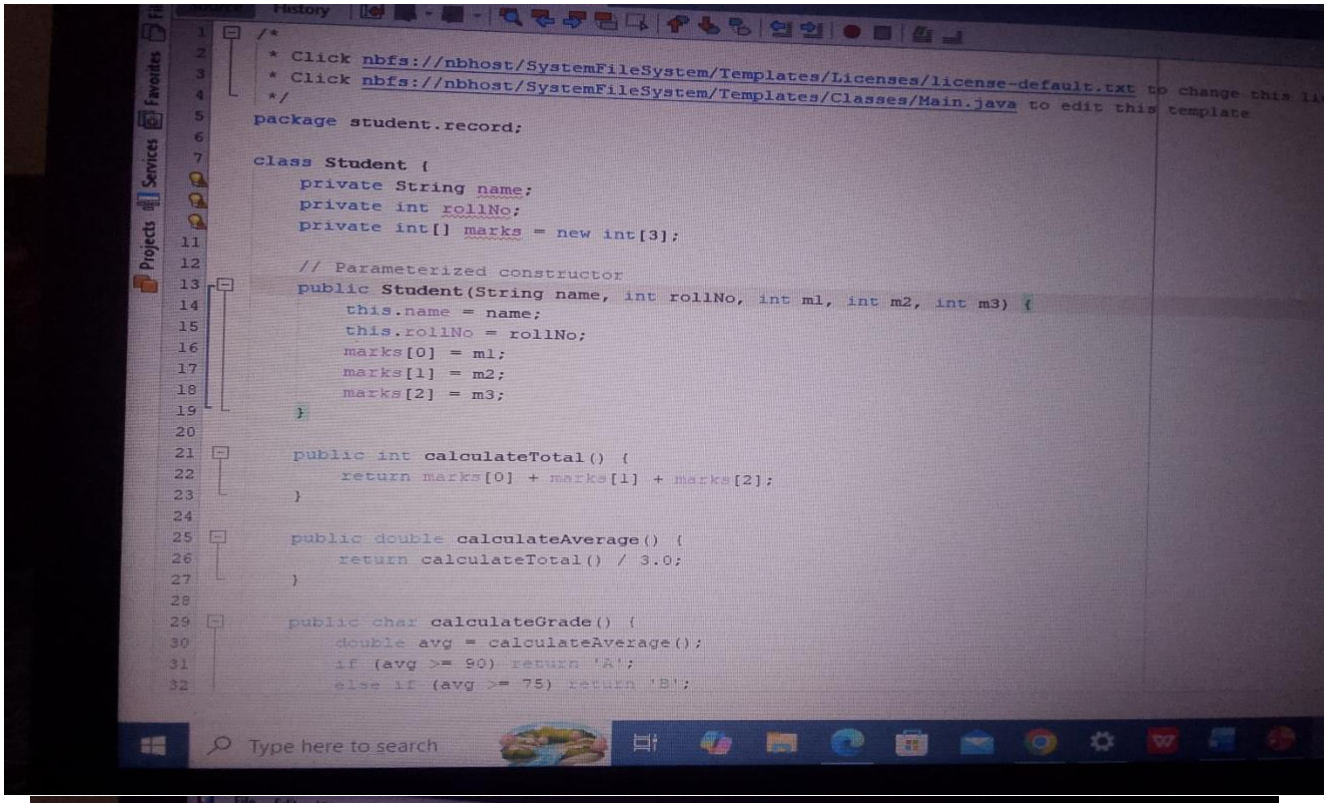
Task 2 — Student Result System

Design a Student Result System using a class named Student. Store private members name, rollNo and marks for three subjects. Use a parameterized constructor and member functions to calculate total, average and grade.

Create an array of at least three Student objects and display the result of every student.

Requirements:

- Use private data members.
- Use a constructor to initialize each student.
- Use separate functions for total and average.
- Assign a grade based on average.
- Use an array of objects.
- Display a neat result for all students.



```java
/*
 * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this li
 * Click nbfs://nbhost/SystemFileSystem/Templates/Classes/Main.java to edit this template
 */
package student.record;

class Student {
    private String name;
    private int rollNo;
    private int[] marks = new int[3];

    // Parameterized constructor
    public Student(String name, int rollNo, int m1, int m2, int m3) {
        this.name = name;
        this.rollNo = rollNo;
        marks[0] = m1;
        marks[1] = m2;
        marks[2] = m3;
    }

    public int calculateTotal() {
        return marks[0] + marks[1] + marks[2];
    }

    public double calculateAverage() {
        return calculateTotal() / 3.0;
    }

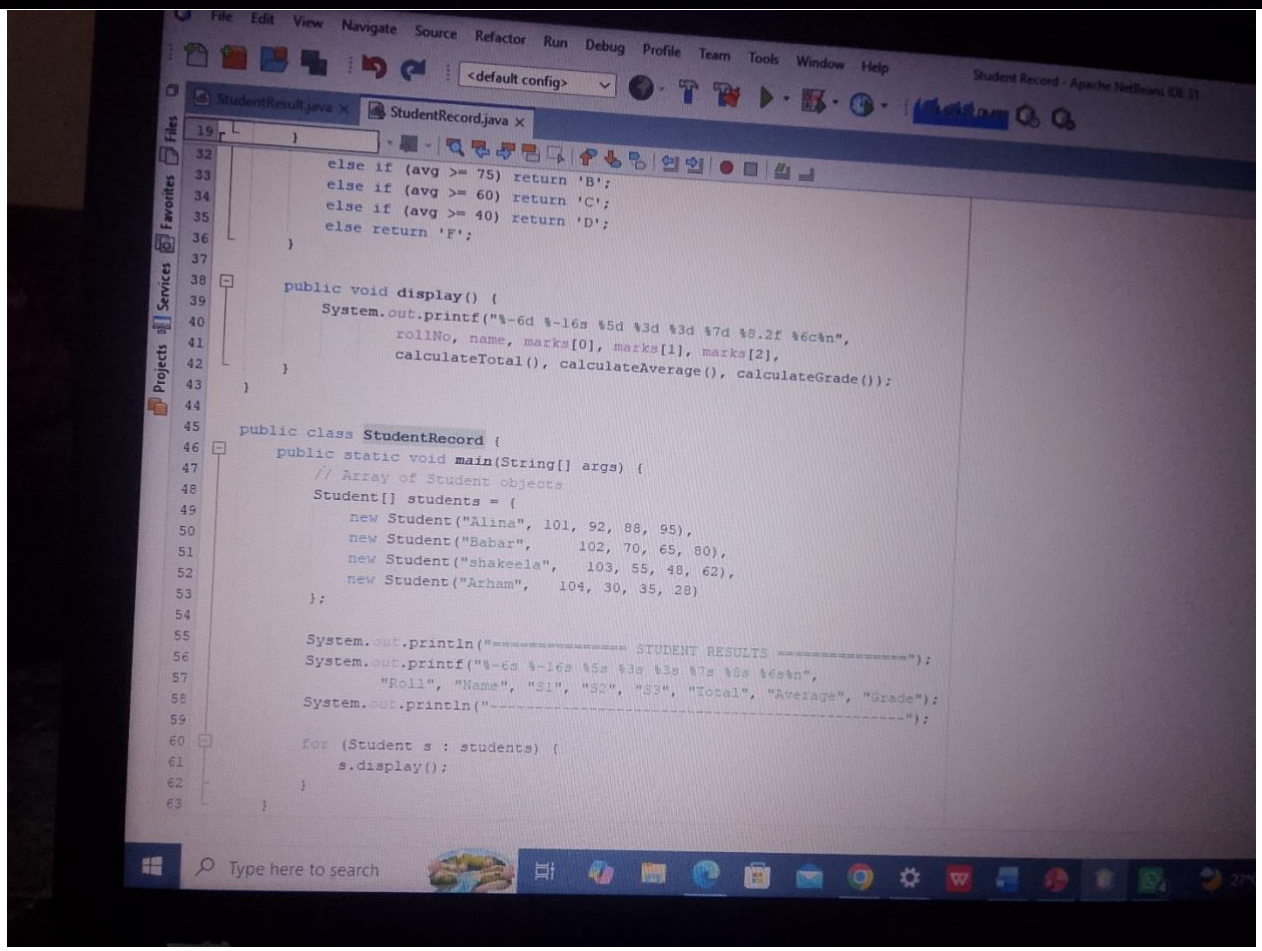
    public char calculateGrade() {
        double avg = calculateAverage();
        if (avg >= 90) return 'A';
        else if (avg >= 75) return 'B';
        else if (avg >= 60) return 'C';
        else if (avg >= 40) return 'D';
        else return 'F';
    }

    public void display() {
        System.out.printf("%-6d %-16s %5d %3d %3d %7d %8.2f %6c%n",
                rollNo, name, marks[0], marks[1], marks[2],
                calculateTotal(), calculateAverage(), calculateGrade());
    }
}

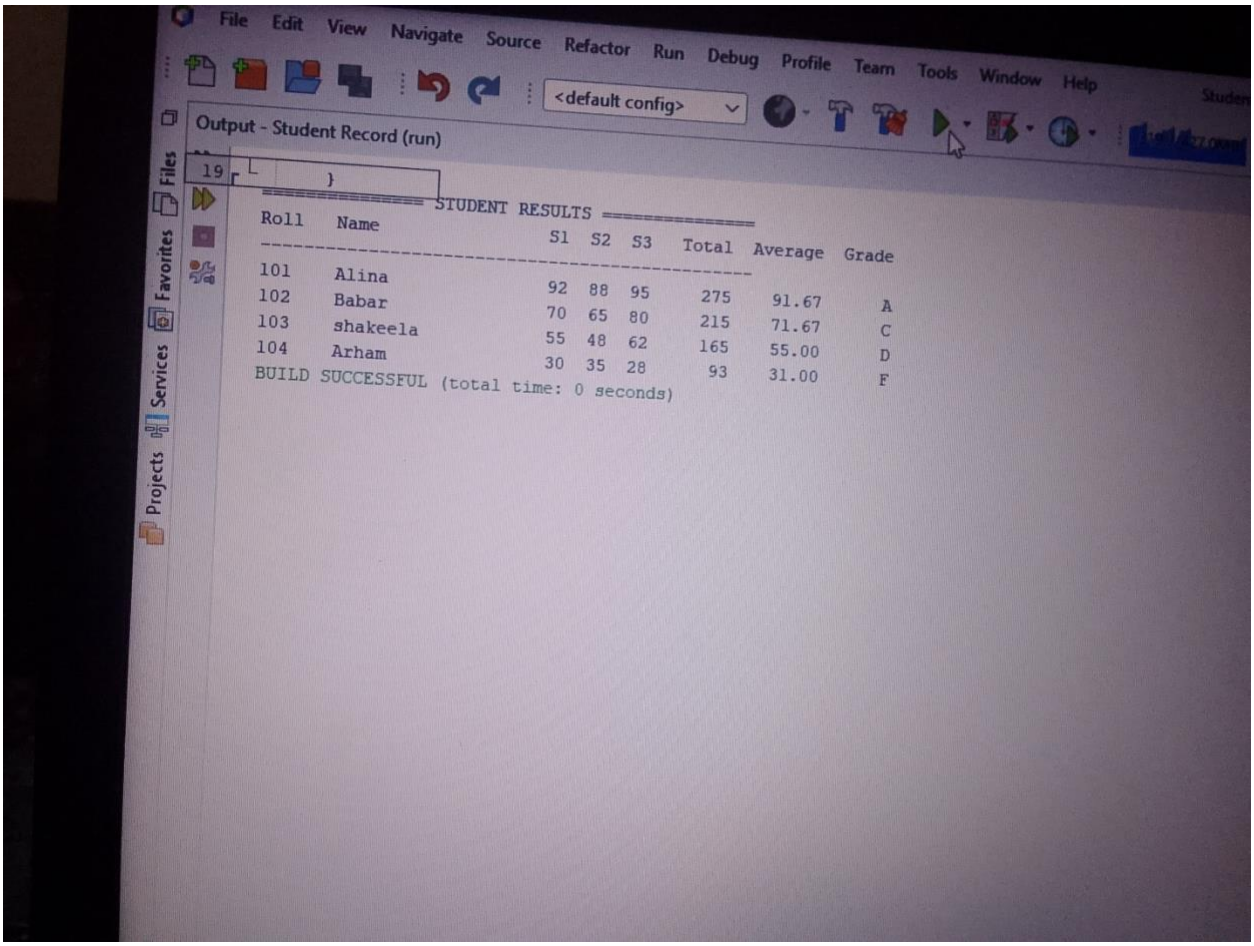
public class StudentRecord {
    public static void main(String[] args) {
        // Array of Student objects
        Student[] students = {
            new Student("Alina", 101, 92, 88, 95),
            new Student("Babar",     102, 70, 65, 80),
            new Student("shakeela",  103, 55, 48, 62),
            new Student("Arham",   104, 30, 35, 28)
        };

        System.out.println("================ STUDENT RESULTS ================");
        System.out.printf("%-6s %-16s %5s %3s %3s %7s %8s %6s%n",
                "Roll", "Name", "S1", "S2", "S3", "Total", "Average", "Grade");
        System.out.println("-------------------------------------------------");

        for (Student s : students) {
            s.display();
        }
    }
}
```

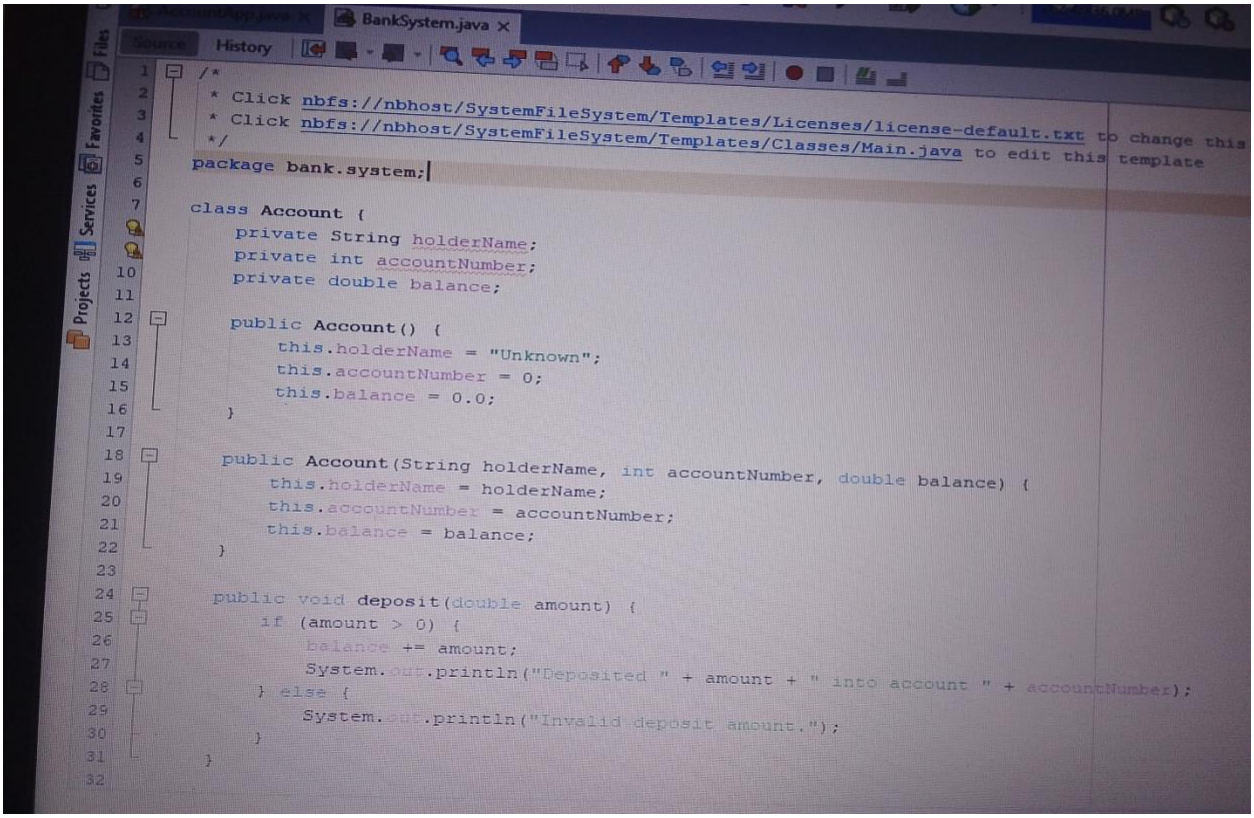


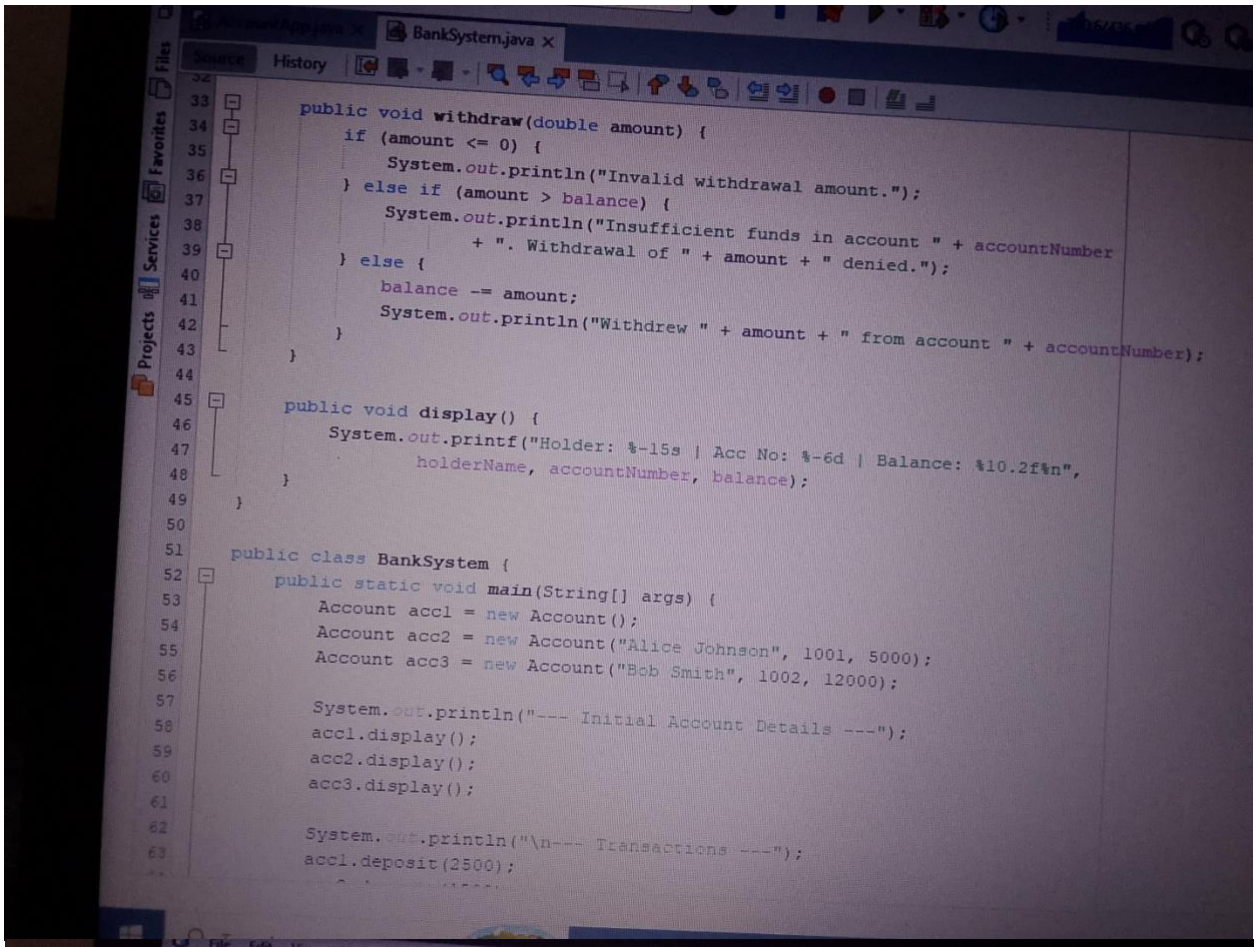
Output:


File Edit View Navigate Source Refactor Run Debug Profile Team Tools Window Help
<default config>
Output - Student Record (run)
Files Favorites Services Projects
19 }
STUDENT RESULTS
Roll Name S1 S2 S3 Total Average Grade
101 Alina 92 88 95 275 91.67 A
102 Babar 70 65 80 215 71.67 C
103 shakeela 55 48 62 165 55.00 D
104 Arham 30 35 28 93 31.00 F
BUILD SUCCESSFUL (total time: 0 seconds)

Task 3 — Bank Account Management System

- Create an Account class with:
- Account holder name, account number, and balance.
- Use appropriate access specifies.
- Implement a default constructor and a parameterized constructor.
- Create objects for 3 accounts.
- Create functions to deposit, withdraw, and display account information.
- Display the final balance of each account.





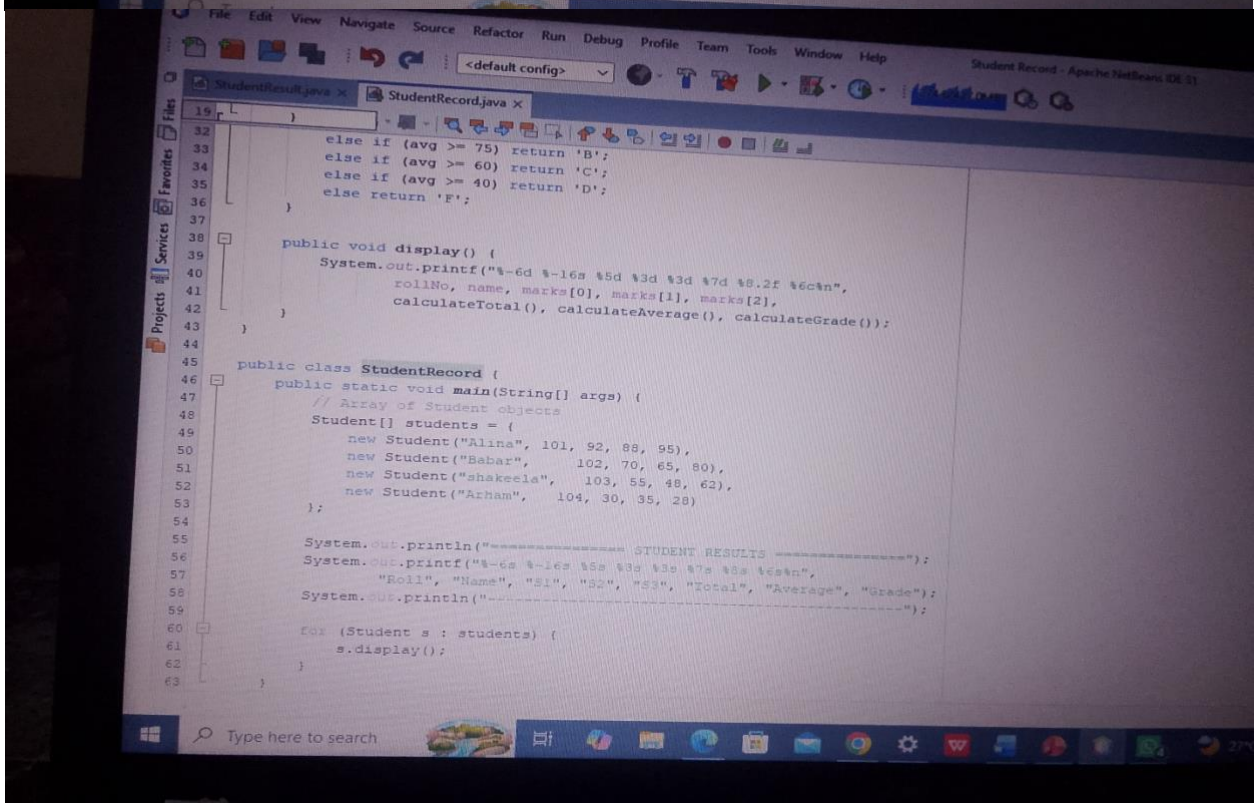
```java
    public void withdraw(double amount) {
        if (amount <= 0) {
            System.out.println("Invalid withdrawal amount.");
        } else if (amount > balance) {
            System.out.println("Insufficient funds in account " + accountNumber
                    + ". Withdrawal of " + amount + " denied.");
        } else {
            balance -= amount;
            System.out.println("Withdrew " + amount + " from account " + accountNumber);
        }
    }

    public void display() {
        System.out.printf("Holder: %-15s | Acc No: %-6d | Balance: %10.2f%n",
                holderName, accountNumber, balance);
    }
}

public class BankSystem {
    public static void main(String[] args) {
        Account acc1 = new Account();
        Account acc2 = new Account("Alice Johnson", 1001, 5000);
        Account acc3 = new Account("Bob Smith", 1002, 12000);

        System.out.println("--- Initial Account Details ---");
        acc1.display();
        acc2.display();
        acc3.display();

        System.out.println("\n--- Transactions ---");
        acc1.deposit(2500);
```



```java
        else if (avg >= 75) return 'B';
        else if (avg >= 60) return 'C';
        else if (avg >= 40) return 'D';
        else return 'F';
    }

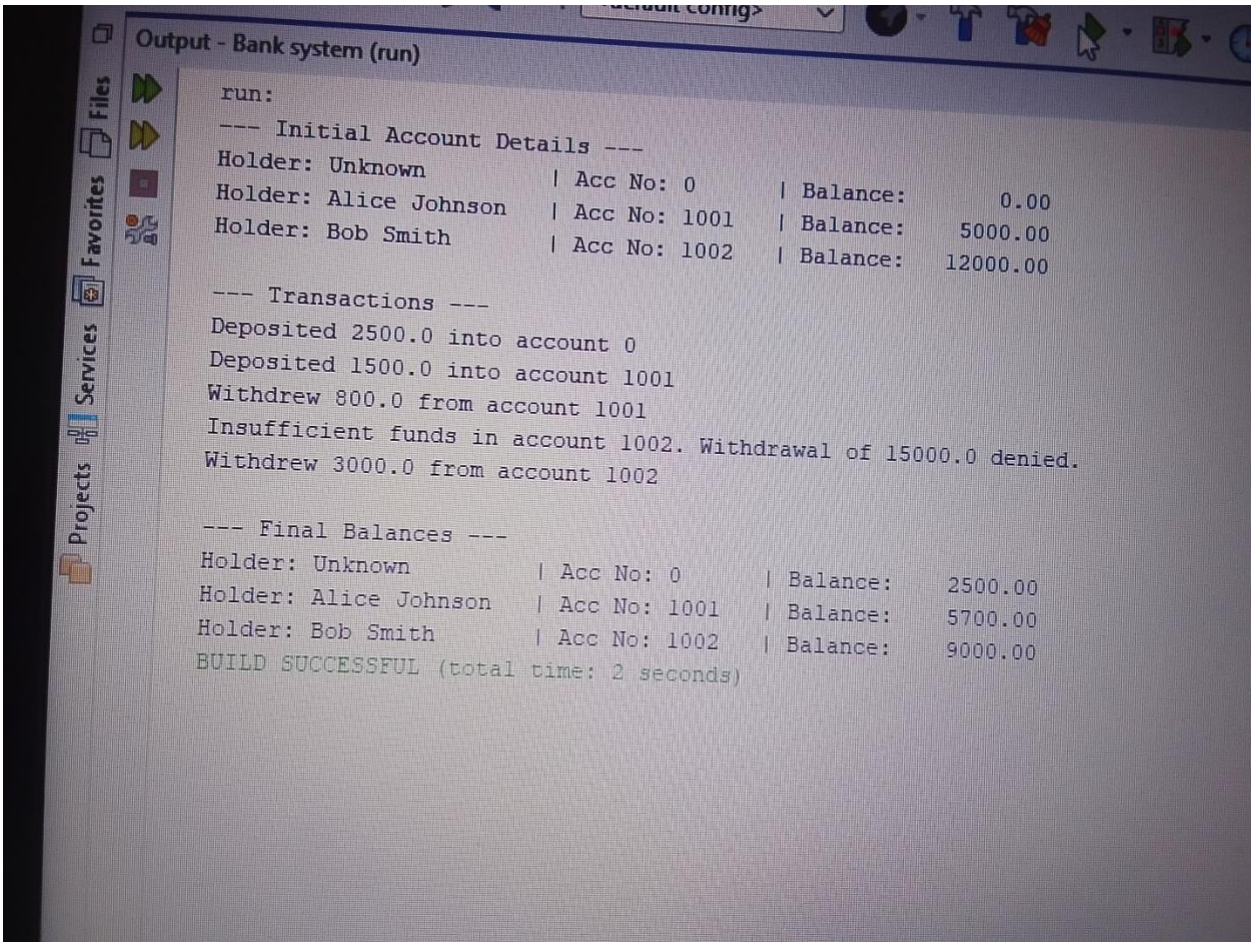
    public void display() {
        System.out.printf("%-6d %-16s %5d %3d %3d %7d %8.2f %6c%n",
                rollNo, name, marks[0], marks[1], marks[2],
                calculateTotal(), calculateAverage(), calculateGrade());
    }
}

public class StudentRecord {
    public static void main(String[] args) {
        // Array of Student objects
        Student[] students = {
            new Student("Alina", 101, 92, 88, 95),
            new Student("Babar",     102, 70, 65, 80),
            new Student("shakeela",  103, 55, 48, 62),
            new Student("Arham",     104, 30, 35, 28)
        };

        System.out.println("========== STUDENT RESULTS ==========");
        System.out.printf("%-6s %-16s %5s %3s %3s %7s %8s %6s%n",
                "Roll", "Name", "S1", "S2", "S3", "Total", "Average", "Grade");
        System.out.println("-----------------------------------------------------------");

        for (Student s : students) {
            s.display();
        }
    }
```

Output:


Output - Bank system (run)
run:
--- Initial Account Details ---
Holder: Unknown | Acc No: 0 | Balance: 0.00
Holder: Alice Johnson | Acc No: 1001 | Balance: 5000.00
Holder: Bob Smith | Acc No: 1002 | Balance: 12000.00
--- Transactions ---
Deposited 2500.0 into account 0
Deposited 1500.0 into account 1001
Withdrew 800.0 from account 1001
Insufficient funds in account 1002. Withdrawal of 15000.0 denied.
Withdrew 3000.0 from account 1002
--- Final Balances ---
Holder: Unknown | Acc No: 0 | Balance: 2500.00
Holder: Alice Johnson | Acc No: 1001 | Balance: 5700.00
Holder: Bob Smith | Acc No: 1002 | Balance: 9000.00
BUILD SUCCESSFUL (total time: 2 seconds)
Files
Favorites
Services
Projects